

Short summary/Objective: Designing a compact, reliable maritime perception stack by: (i) using classical and temporal object tracking on radar imagery; (ii) learning features with self-supervised temporal masked autoencoding; (iii) aligning multi-spectral sensors (RGB/thermal) for fusion.

Summary: Using frequency-aware filtering and land-masking to stabilize radar frames; utilizing intensity, background-subtraction, and motion cues with gap-tolerant temporal association to produce persistent vessel tracks; learning features with temporal masked autoencoding (frame/patch masking, cross-gain prediction) to improve robustness under low SNR; and combining calibrated multi-spectral alignment (RGB/thermal) via emissivity-contrast targets and MI/phase-based registration to enable downstream fusion.

Sea-ing in Binary: Learning without Teachers & Self-Supervisions

Part 1: Overview of Outcomes

Accurate maritime perception from a compact X-band radar demands reconciling low-dynamic-range, noise-laden echoes with the precision required for collision avoidance, search-and-rescue, and autonomous navigation. Over the reporting interval, four parallel efforts addressed this gap.

In this period we: (1) implemented end-to-end blob classification and temporal persistence tracking; (2) trained temporal MAE variants with frame/patch masking, producing reconstruction grids and PSNR/SSIM curves; (3) and evaluated multi-spectral registration pipelines on sample pairs and summarized quantitative metrics.

Part 2: Radar Object Tracking

This section summarizes what the figures show and what we learned while building an end-to-end tracker for small vessels in coastal radar. The pipeline progressed from frequency-domain filtering (to suppress near-zero Doppler clutter) into three complementary detectors—intensity, background subtraction, and frame-difference motion—and finally into temporal association that turns per-frame blobs into tracks. Rotation alignment was applied from per-file JSON metadata so that the coastline mask and the video frames share a common azimuth/range frame; mis-registration here is the main source of shoreline “bleed.”

The detector layer begins with a conservative blob extractor: grayscale thresholding in the radar intensity domain, followed by connected-component contours and an area range filter to drop speckle and land fragments. A hard cap on blob count prevents storms of false alarms during adverse frames; any very large region is force-typed as land. A land/water mask then classifies detections via centroid lookup; we keep HSV color overlays strictly for debugging to make it obvious when mask edges are off. Temporal association uses spatial hashing with cell sizes

that adapt with $\sqrt{\text{area}}$ so that larger objects tolerate bigger displacements. We allow short gaps in the data to fill missing detections and reduce flicker. Each track accumulates a lifetime; a presence threshold controls the labeling scheme—long-lived tracks render as persistent (purple), while short-lived, flickery tracks render as transient (red). The system emits per-frame overlays, JSON/TXT summaries, and stitched videos for review.

Figure 1 combines four diagnostics. The background model (top-left) isolates static land and bright shoreline artifacts; this is the baseline we subtract from. The track overlay (top-right) shows the top eleven tracks meandering near islands and coast—the concentration of colored polylines near land hints that residual mask error or sidelobe energy is still leaking into the water. The objects-per-frame plot (bottom-left) oscillates between roughly 8 and 20 candidates with spikes above 22, which is too many for a scene that rarely holds that many boats; these spikes correlate with sea-state shimmer and mask misalignment. The track-length histogram (bottom-right) is sharply skewed: many tracks die within a few dozen detections, but a handful persist hundreds of frames. That long tail is good—it means the pipeline can sustain real targets—but the large mass of short tracks indicates fragmentation from clutter and occasional association breakage.

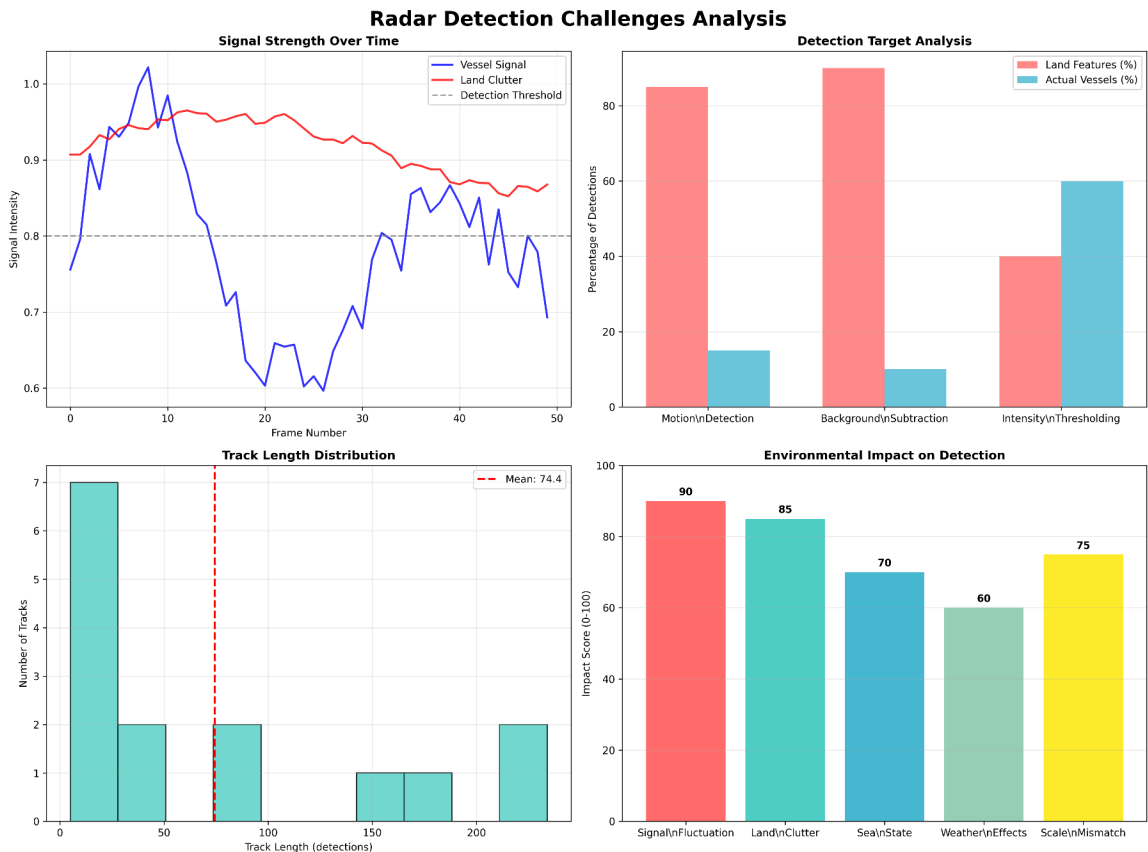


Figure 1: Review of methods and results

To understand why simple detectors struggle, we looked in the Fourier domain. Figure 2 shows 2-D FFT slices and a maximum-intensity projection. Energy piles up along the horizontal and vertical axes around the origin—the signature of near-zero Doppler clutter and structural edges in

range/azimuth. This explains why intensity thresholds fire on coastlines and why motion differencing lights up everywhere under small mis-registrations. The implication is practical: apply high-pass or notch filters around zero frequency, or equivalently, emphasize temporal change after careful stabilization and background modeling.

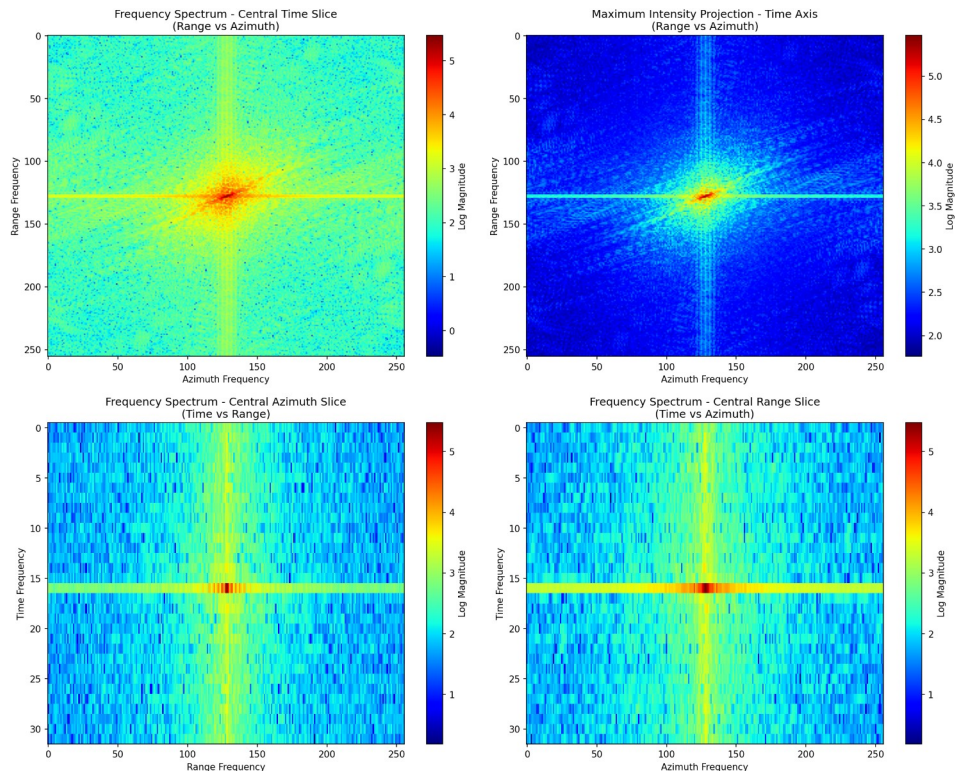


Figure 2: 2D FFT of radar frames across different dimensions

Figure 3 illustrates intensity-based tracking. Bright vessels are reliably picked up and, once the land mask suppresses most shoreline clutter, tracks persist with stable IDs. The downside is precision near land: small mask errors cause spurious blobs, and wake fragments can trigger multiple short tracks. We mitigate this with morphology (to merge adjacent blobs), stricter area/elongation constraints, and higher presence thresholds near the shoreline.

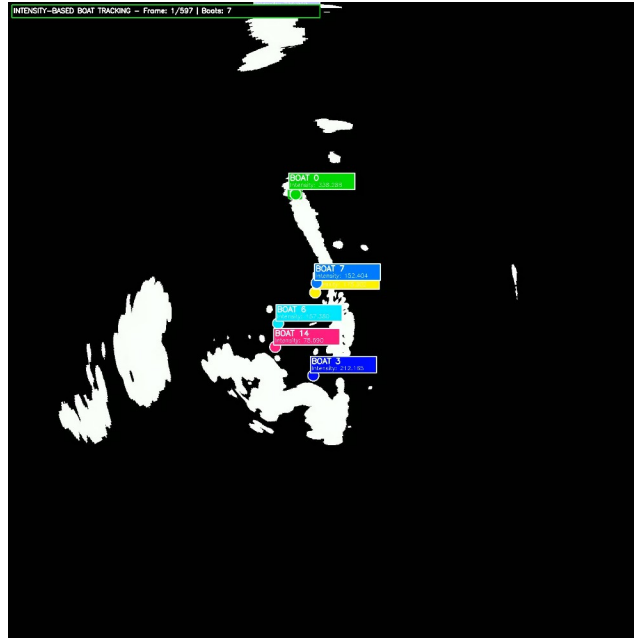


Figure 3: Intensity Based Detections and Tracking

Figure 4 demonstrates background subtraction. The middle panel shows the absolute difference from the learned background; the right panel thresholds at the 95th percentile, yielding a compact set of candidates clustered away from land. Compared with pure intensity, background subtraction is less sensitive to global gain and better at surfacing dim, moving targets, but it still reacts to slow background drift and can miss quasi-stationary vessels. In practice we run it alongside intensity and keep only cross-confirmed detections or those that mature into persistent tracks.

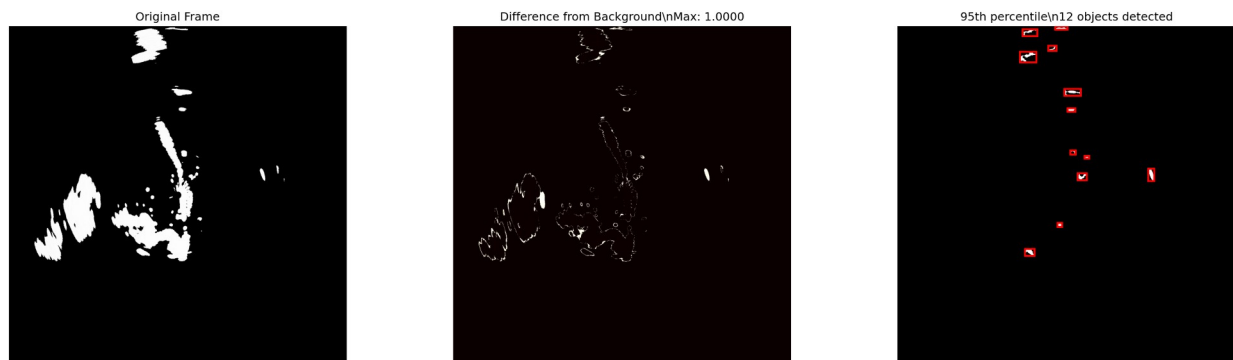


Figure 4: Background Subtraction Detection Method

Finally, Figure 5 summarizes diagnostic trends. Vessel signal often sits below land clutter; without masking, motion and background models overwhelmingly “detect” land. Track lengths cluster short unless detectors are fused and association is allowed to bridge gaps. Environmental factors—signal fluctuation and land clutter foremost, followed by sea state and scale mismatch—drive most errors. The takeaway is that fusion beats any single cue: frequency-aware filtering to suppress low-Doppler clutter, intensity for bright targets, background subtraction for robustness to gain, conservative motion cues for confirmation, and a gap-tolerant association layer.

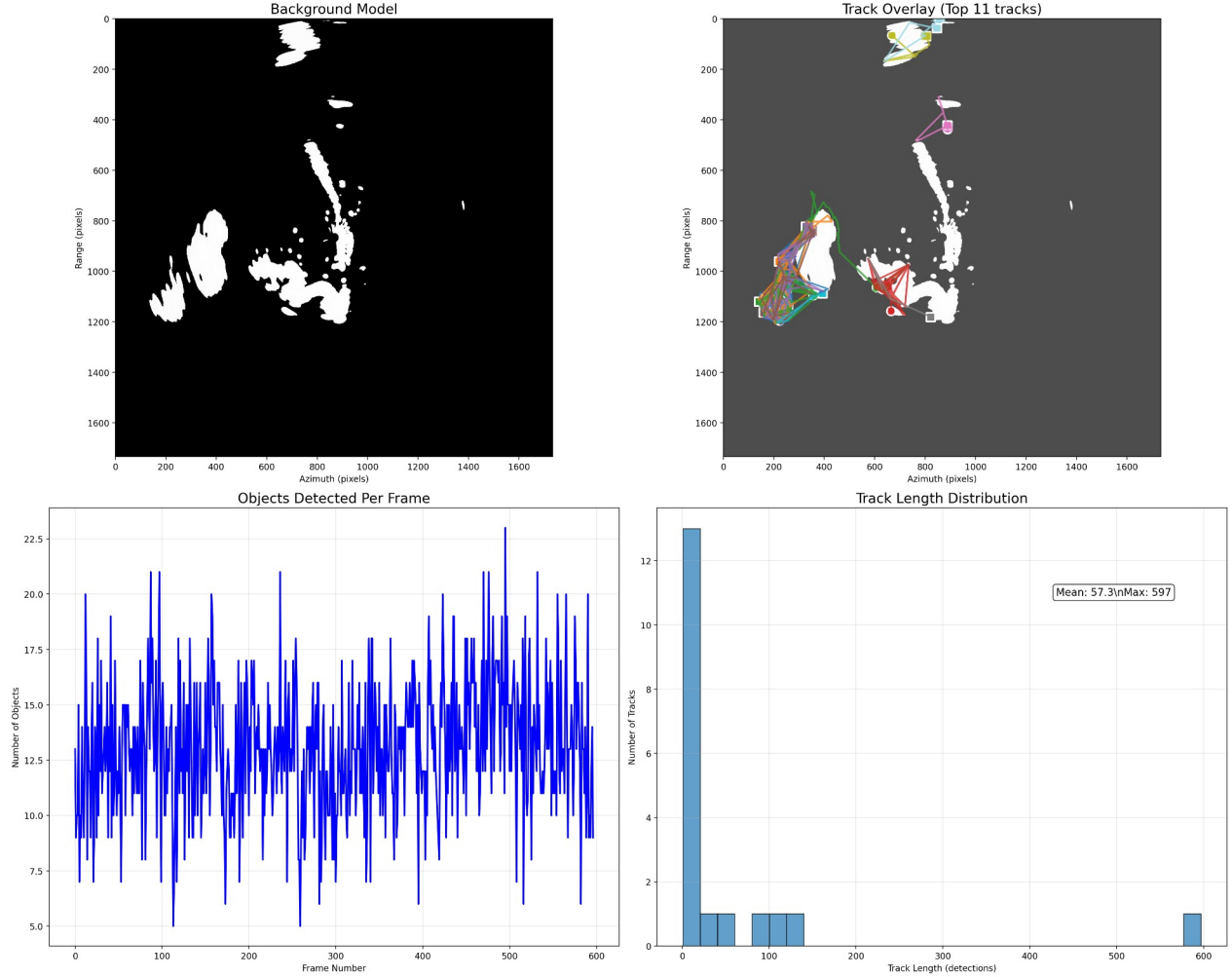


Figure 5: Review of background subtraction model

Part 3: Self-Supervised Machine Learning

Continuing with the proposed idea of using self-supervised masked auto encoding to learn features of radar videos, then using a few labeled snapshots to create a lightweight detection head, during this reporting period, multiple variants of were tested and characterized.

We begin with a standard MAE system, but discover severe limitations of existing models, therefore develop our own custom architectures. Following this, the models are tested on the CIFAR-10 dataset with varying mask ratios and model sizes, allowing for comparison to standard masked autoencoder models from the likes of Google and Meta.

Across seven model sizes (small through gigantic), the encoder scales from 384 to 2048 embedding dimensions with depth and attention heads increasing proportionally; each is paired with a smaller decoder (for example, the “large” setting uses a 768-dim, 12-block, 12-head encoder and a 512-dim, 8-block, 16-head decoder). Defaults for a typical run are modest: 32×32 inputs cut into 4×4 patches with 3 channels, AdamW as the optimizer (weight decay 0.05, betas 0.9/0.999, eps $1e-8$), an initial learning rate of $1e-4$, 80 training epochs, cosine LR scheduling, and periodic checkpointing and visualizations every five epochs. Alternative optimizers (Adam,

SGD, RMSprop) and schedulers (linear, step, multistep, exponential, plateau, warmup-cosine, polynomial, or none) are available when different dynamics are desired.

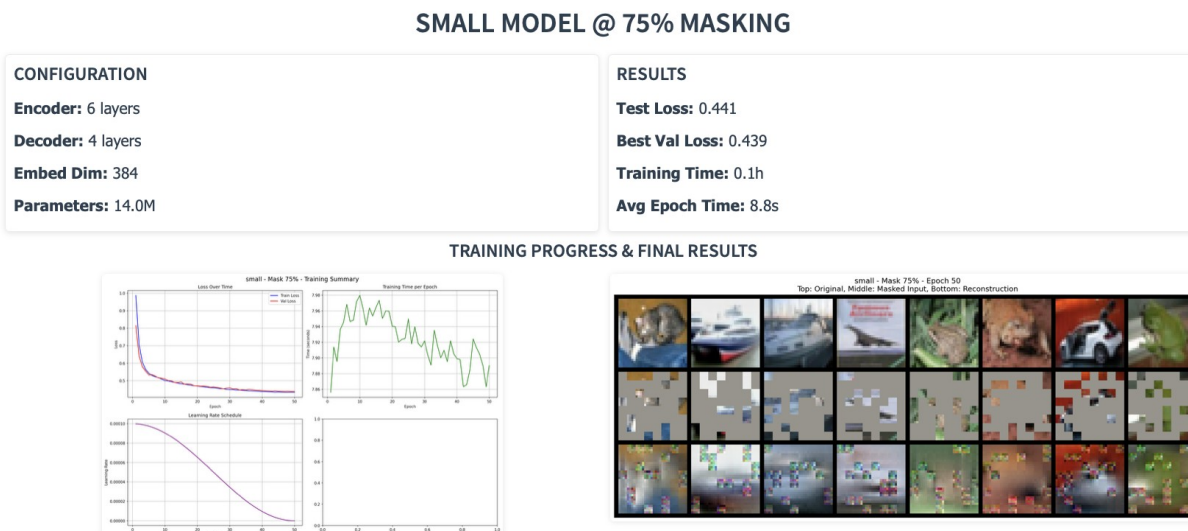


Figure 6: Custom MAE results on CIFAR-10 data

A systematic experiment sweep shows that the masking ratio is the single most influential hyperparameter. Lower masking—around 35–60%—consistently outperforms the canonical 75% and dramatically beats very high masking. Representative results for the large model are test losses of 0.189 at 35%, 0.239 at 50%, 0.296 at 60%, rising to 0.414 at 75% and 0.709 at 90%. Medium models achieve similar quality at a fraction of the parameters and time—for example, 0.193 at 35% in roughly 0.4 hours with 36.2M parameters—indicating a strong accuracy-efficiency trade-off and diminishing returns from brute-force scaling. In total, fifteen runs covered the size–masking grid to support these conclusions.

Scaling still helps, but the gains are sub-linear: doubling parameters does not double performance. As a result, it is usually better to tune masking and training schedules first, then increase model capacity judiciously. Adaptive training practices—cosine or warmup-cosine schedules and early stopping—improve stability and reduce wasted compute.

For qualitative evaluation, a hybrid reconstruction display is recommended: render original pixels for the visible patches and show model predictions only for the masked ones. This avoids the misleading, garbled appearance of standard MAE visualizations where even visible patches are reconstructed, and it makes it easier to judge what the model actually predicted. In practice, the recipe that emerges is simple: start with a medium model, set masking below 75% (often 35–60%), use AdamW with cosine scheduling, enable early stopping and clear hybrid visualizations, and only then consider scaling up the encoder if the task demands higher capacity. Results from a small model is shown in Figure 6.

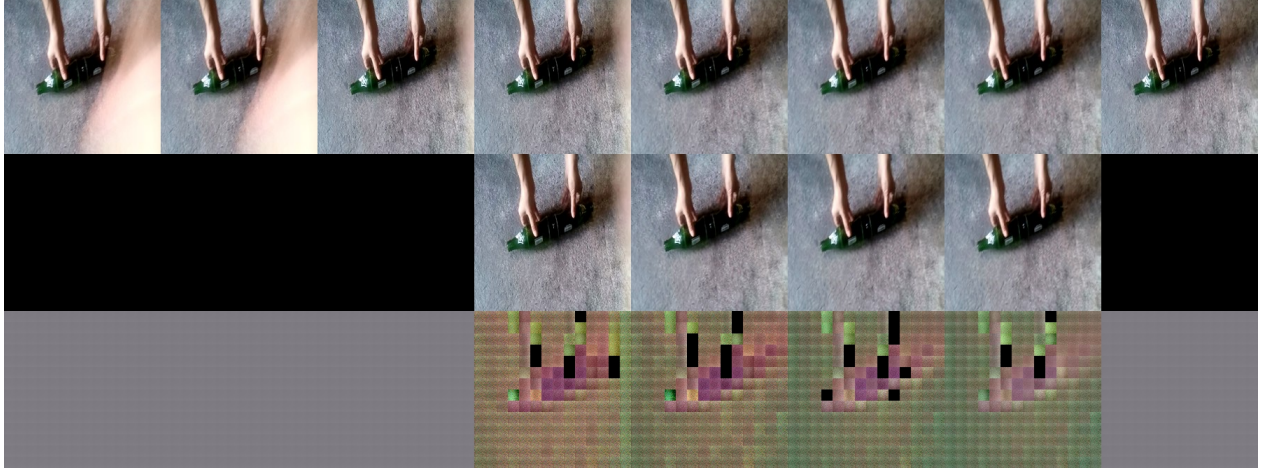


Figure 7: Temporal MAE on the Something-Something v2 dataset, epoch 50

We developed a temporal Masked Autoencoder (T-MAE) on top of the earlier MAE framework. Unlike the first round of MAE experiments, this version was trained end-to-end on a MacBook, which confirmed that the method is portable and can be iterated on without a large compute cluster. The objective remains reconstruction of masked content, but the masks now evolve over time so the model must leverage temporal context as well as spatial appearance.

To probe what the model actually learns, we compared several masking strategies. Full-frame masking removes contiguous frame regions, encouraging long-range context use but sometimes making the task too easy by hiding static backgrounds wholesale. Random patch masking samples spatial patches independently across time, producing diverse missing regions that couple motion cues with appearance. Static patch masking fixes the patch pattern across frames; because the same locations are hidden at each timestep, the model is pushed to reason about temporal dynamics inside the visible areas instead of relying on fresh pixels appearing later. These three strategies are illustrated in Figure 8.

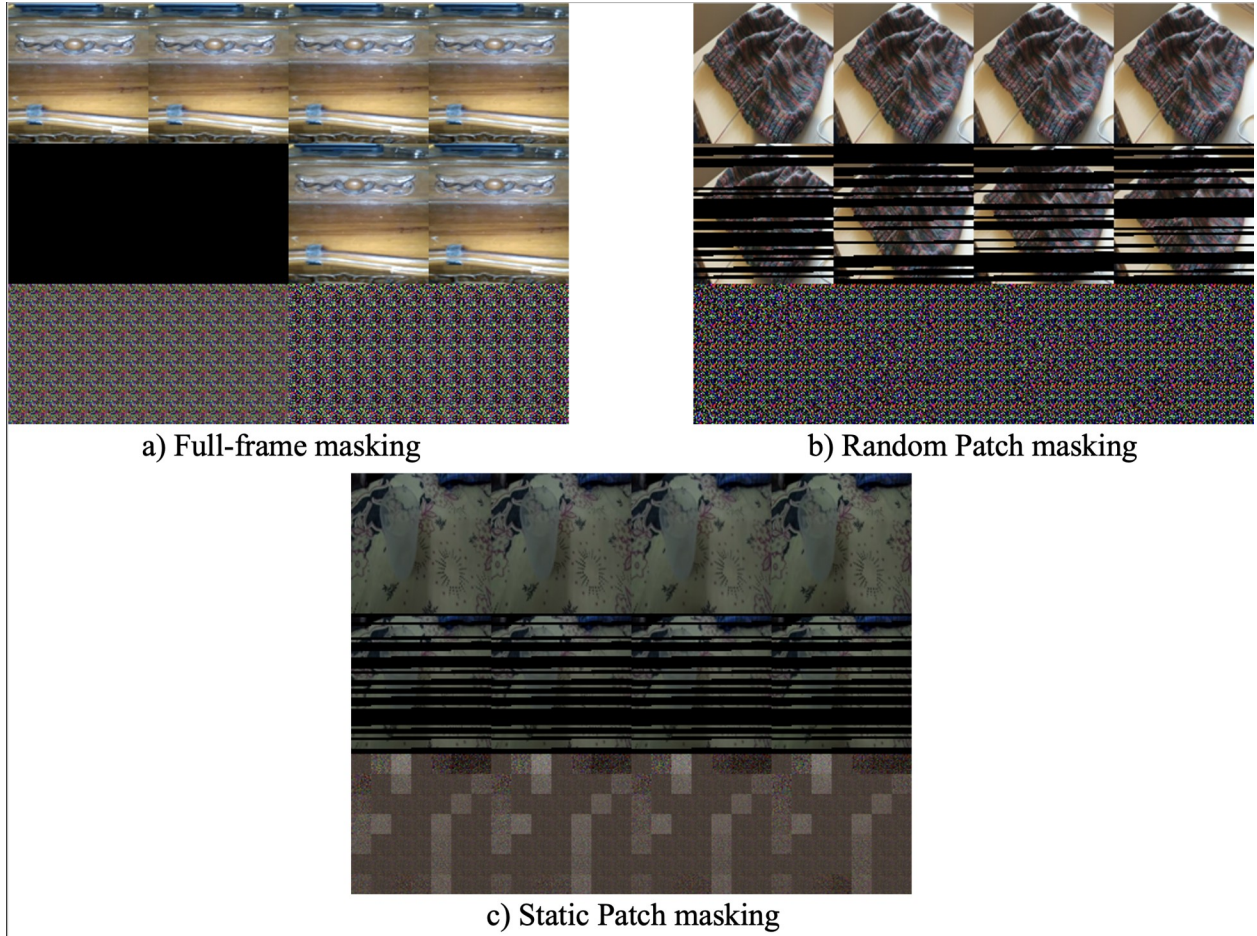


Figure 8: Different Masking methods for Temporal Masked Autoencoding

On a standard benchmark (Something-Something V2), the temporal MAE produces coherent reconstructions after 50 epochs and shows steady loss improvements, suggesting the temporal signal is being captured rather than merely copying frame-to-frame textures. Figure 7 summarizes qualitative results at epoch 50.

We then adapted the same frame-erasure objective to a radar setting: given a higher-gain frame (e.g., gain 70), predict what a lower-gain version (gains 40 and 50) would look like. Early results are promising—the model reconstructs plausible low-gain views from high-gain context—but questions remain about how much of the useful feature content is preserved for downstream tasks like vessel detection. To answer this, we plan ablations with perceptual and task-based metrics, cross-gain cycle-consistency checks, and training with mixed-gain sequences.

Finally, we extend temporal masking in both directions—predicting low-gain from high-gain context and high-gain from low-gain—on synthetic and real sequences. We include heat-map differences and selected real samples to illustrate the task difficulty and the learned improvements. Figure 9 shows representative predictions that qualitatively support the approach.

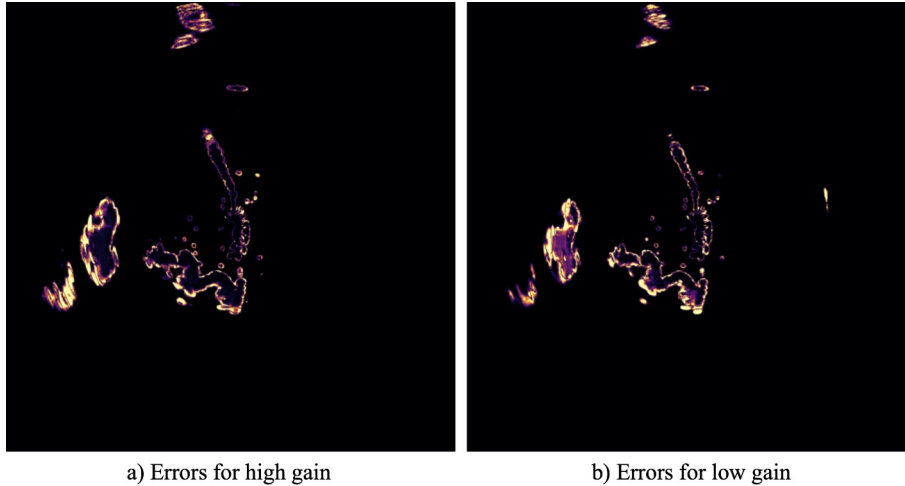


Figure 9: Results of predictions for low and medium gain from a high gain image

We also explored whether zero-shot methods could help with radar by first validating them on standard natural-image datasets. The prototype used a text-promptable detector to propose boxes (e.g., Grounding-DINO-style phrase queries such as “wolf”, “sheep”, “log”), fed those boxes to a promptable segmenter (SAM/SAM-like) to obtain instance masks, and then assigned labels with a zero-shot classifier (CLIP-style image–text similarity over a small label list and synonyms). The figure below shows a typical result: starting from the original image, the system draws high-recall boxes around candidate objects, converts each box into a binary mask, and overlays the segmentation. We also record mask statistics (min/max logits, non-zero pixel counts) and, for visualization, threshold the raw mask to highlight the foreground region. An example is shown in Figure 10.

Mechanically, the pipeline works as follows. A short list of text prompts is tokenized and passed to the open-vocabulary detector, which returns boxes with per-phrase scores; non-max suppression merges overlaps while preserving recall. For each surviving box, we call the segmenter in “box-prompt” mode to generate per-instance masks and logits. We post-process masks with a fixed or percentile-based threshold, optional morphological smoothing, and confidence gating that combines detector and mask scores. For labelling, we crop masked regions, compute image–text similarity to the prompt set, and keep the top label if the margin over the runner-up exceeds a threshold; otherwise the instance is marked “unknown.” Everything runs frame-by-frame, which kept the prototype simple but introduced temporal flicker when extended to video.

Empirically, results on natural images were acceptable for salient objects, but performance quickly degraded for small, camouflaged, or fine-grained categories, and the segmentation quality depended heavily on the phrasing of prompts and the detector’s calibration. More importantly, we found no robust, public method that extracted zero-shot video masks end-to-end in a way that transferred cleanly to our data at the time of testing. When we tried to apply the approach to radar, domain shift became the primary blocker: radar targets lack stable natural-image semantics, CLIP-style text supervision is poorly aligned with radar appearance, and the per-frame pipeline produced inconsistent masks across time. Given these issues, and no clear path to quick wins, we paused this line of work.

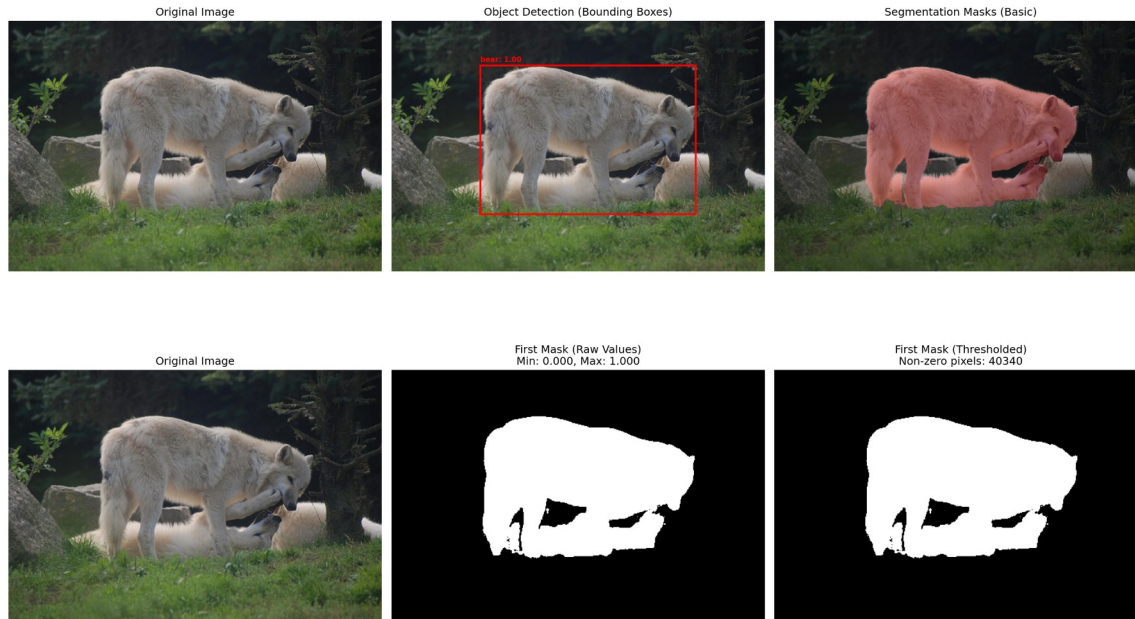


Figure 10: Zero-shot classification, identification and segmentation

That said, the idea remains promising as an onboarding project. A sensible next pass would add a small, curated radar label set for prompt tuning; adapt the segmenter with lightweight finetuning (e.g., LoRA) on radar patches; incorporate motion cues to stabilize masks across frames; and explore open-vocabulary video variants to enforce temporal consistency. If future pretrained models close the domain gap, revisiting zero-shot radar segmentation could become viable.

Part 4: Multi-Spectral Alignment

We addressed the problem of aligning imagery from heterogeneous sensors—principally RGB, depth, and long-wave infrared (thermal). The goal is to place all signals into a common pixel coordinate system so that downstream models can fuse them cleanly. When co-registered correctly, per-pixel overlays such as depth edges or thermal gradients act as strong auxiliary cues for detection and segmentation. The core difficulty is that cross-spectral pairs rarely share the same appearance: features that are obvious in RGB can be faint or absent in thermal, and standard calibration targets may not respond at all in the infrared band.

Our calibration strategy begins with per-sensor intrinsics and distortion, followed by cross-sensor extrinsics and a final dense image registration. For RGB - RGB we use conventional checkerboard targets, solve intrinsics, and estimate a homography for planar scenes or full extrinsics for general 3-D setups. Thermal requires a different target because a printed checkerboard has uniform temperature and therefore no contrast. We designed a target that produces emissivity-driven contrast: a matte, high-emissivity substrate backed by a controllable heat source with large circular apertures exposing a low-emissivity plate. In thermal, the emissivity contrast yields crisp disks; in RGB, the same geometry is visible as high-contrast circles. The target is robust to modest temperature drift and works indoors without extreme heating. Figure 11 illustrates the CAD concept for this dual-band target.

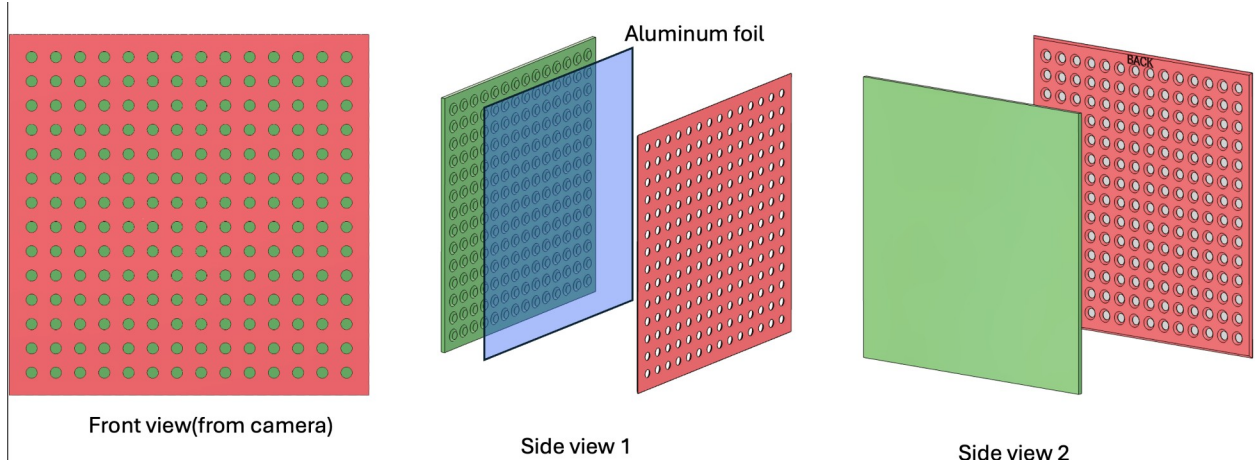


Figure 11: CAD model of new checkerboard for thermal and RGB

Data collection follows a short protocol. Sensors are time-synchronized and mounted rigidly. We capture a grid of views that span the field of view and depth, pausing between shots to let the thermal target settle. Each modality is calibrated independently to obtain intrinsics and lens distortion. We then estimate cross-sensor extrinsics using the circle centers as correspondences, solving for a rigid transform. For planar scenes we compute an initial homography; for general scenes we use the rigid transform to warp into a reference camera and perform a coarse-to-fine image registration.

We evaluated three families of registration criteria on paired frames: direct intensity objectives such as SAD, SSD, normalized cross-correlation, and mutual information (MI); frequency-domain alignment via phase correlation; and feature-based alignment using SIFT, SURF, ORB, and Harris corners. Because appearance differs across spectra, MI and phase correlation were more stable, while purely intensity-based metrics were brittle outside $\text{RGB} \leftrightarrow \text{RGB}$. Feature-based methods worked well for $\text{RGB} \leftrightarrow \text{RGB}$ and sometimes for $\text{RGB} \leftrightarrow \text{thermal}$ when edges were strong, but degraded with low-contrast backgrounds.

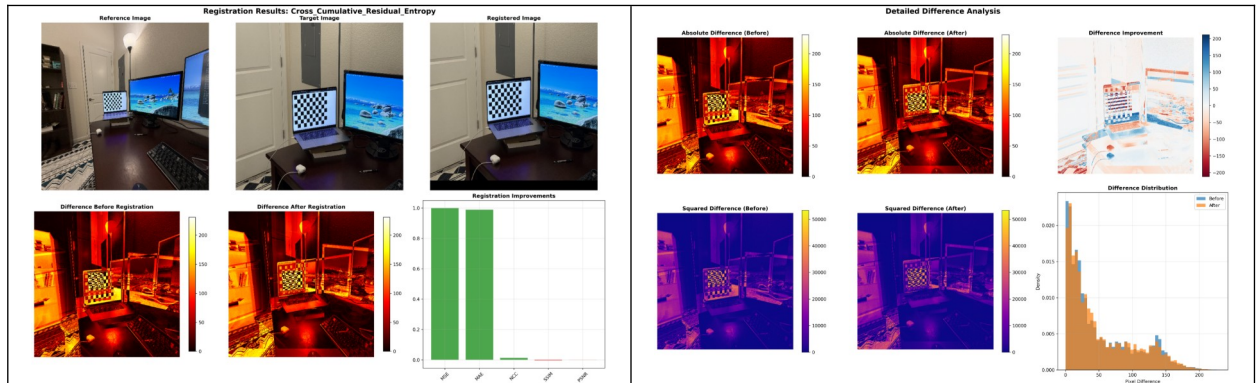


Figure 12: Main comparison and difference maps of RGB to RGB camera alignment

On our comparison set (Figure 12), many methods reduced pixelwise PSNR/SSIM after registration on the cross-spectral pair—consistent with spectral mismatch and small scale/ROI drift. Phase correlation and MI degraded the least, and correlation-coefficient–based registration (CCRE) changed minimally, as expected.

We also compared a generative enhancement baseline (Qwen Image Edit) to a deterministic sharpening/upscaling pipeline tuned for calibration imagery (Figure 13). The generative edit can improve human-perceived clarity but risks altering geometry, which is undesirable for calibration. The deterministic recipe—linear color correction, mild deconvolution, and non-ringing upscaling—preserves edges and tonal neutrality and is therefore preferred before feature detection. Edge density and local entropy metrics support the deterministic choice for our targets.



Figure 13: clarity_contrast. Left: Qwen generative edit (reference). Right: deterministic standard pipeline tuned for edge fidelity and tonal neutrality.

Part 5 Future

Part 5a: Algorithms and Model Development

Train temporal MAE on radar sequences with gain/PRI/sea-state augmentations, then probe with lightweight heads (linear probe or tiny DINO/YOLO-style head) for few-shot detection; explore joint objectives that predict cross-gain and de-cluttered frames so the representation carries denoising priors.

Part 5b: Experiments and Testing

Plan a structured data campaign across the lake using multiple pulse lengths (short/medium/long) and gains (40/50/70), collecting matched sequences under varied sea states, wind, and daylight. For temporal MAE, evaluate reconstruction PSNR/SSIM and downstream detection lift under cross-gain prediction, including cycle-consistency checks (70→40→70).